

A deep learning method using SPECT images to diagnose remaining thyroid tissue post-thyroidectomy

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Abstract—After total thyroidectomy, Single-photon emission computed tomography (SPECT) is used to diagnose whether thyroid tissue remains in patients' bodies. Physicians visually diagnose the residual thyroid tissue in patient base on their expertise, so it is difficult for making a quick and accurate diagnostic. In present, Computer-aided Diagnosis systems (CAD) are becoming more widely in medical treatment, thyroid cancer is a potential field where CAD can improve diagnostic accuracy. This paper proposes a novel approach for diagnosing whether residual thyroid tissues remain in patient using thyroid SPECT scintigraphy by fine-tuning pre-trained Deep neural networks. Our proposed method achieved sensitivity of 84.85% and specificity is 89.11%, these results demonstrate that CAD is promising in diagnosing remaining thyroid tissue after total thyroidectomy.

Index Terms—Thyroid cancer, Focal loss, fine-tuning, thyroid SPECT scintigraphy, Deep learning neural network

I. INTRODUCTION

Thyroid cancer is the most common malignancy of the endocrine system. According to Globocan 2018, thyroid cancer ranks 5th among common cancers in Vietnam that seen in women. Differentiated thyroid cancer (DTC) accounts for over 97% of all thyroid cancers including papillary, follicular, and Hurthle cell. The treatment strategy depends on the type of disease, disease stage, and risk of recurrence. The main methods include surgery, adjuvant therapy with ¹³¹I, and inhibitors of Thyroid-Stimulating Hormone (TSH) with a 5-year survival rate of about 83 -98%.

A SPECT (Single photon emission tomography) scan is a type of nuclear imaging test using gamma rays. Scanning with radioactive ¹²³I or ¹³¹I plays an important role and is considered the "gold standard" in postoperative follow-up [1]. Doctors use SPECT images to assess the extent and distribution of the remaining thyroid tissue after surgery, from which to propose a method appropriate treatment.

Nuclear physician will read the SPECT images and write a report using the image standards like count

rate grayscale, but these standards are not clear, and the diagnosis will depend on the doctor's level of experience. This is one of the disadvantages of the thyroid scan. Diagnostic systems Computer Aided Diagnosis (CAD) are increasingly used to reduce the workload of doctors in current medical practice. On the other hand, avoiding potential medical malpractice claims may require a thorough diagnostic procedure.

More and more CAD systems are being implemented to improve the diagnostic accuracy of various diseases. The scintigraphy of patients with thyroid cancer is a relatively new imaging field for CAD applications with great potential.

In this study, we propose a method for classifying and predicting remaining thyroid tissue using SPECT scintigraphy by employing a deep learning network to address these constraints. In challenging instances, the model can make predictions with less information than a doctor, but it can still do so, that decreases the amount of effort required by physicians.

II. RELATE WORKS

Deep neural network-based CADs generally consist of three sequential stages: pre-processing, feature extraction, and inference. At the time of writing this research, several prior studies have demonstrated that Deep neural networks may be employed as an useful way to help clinical diagnostics.

Using ^{99m}Tc-pertechnetate thyroid scintigraphy, Yang et al. [2] developed an artificial intelligence system based on the Resnet50 model for distinguishing the causes of thyrotoxicosis in healthcare situations. The dataset consists of 2,468 thyroid scintigraphy pictures obtained from West China Hospital for training purposes, as well as 200 testing cohorts from two centers. The researchers next investigated the diagnostic accuracy of four types of thyrotoxicosis: "diffusely increased", "diffusely decreased", "focal increased" and "heterogeneous uptake". Their AI model improves

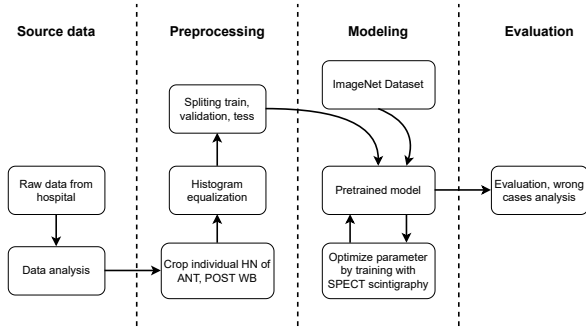


Fig. 1: Pipeline of our proposed method. Our end-to-end solution consists of four sequential stages, each of which involves multiple processes.

diagnosis and consistency than 5 physicians for interpreting data from West China Hospital (accuracy of 88% vs. 66.73%) and Panzihua Central Hospital (accuracy of 83% vs. 53.79%) respectively.

Qiao et al. [3] constructed Deep learning model to diagnose thyroid disease by thyroid scintigraphy. The dataset comprises 1,430 samples from patients with no thyroid disease, patients with Graves disease, patients with subacute thyroiditis during hyperthyroid status, 70% of the samples are designated for training and 30% for testing. They attained accuracy of 91%, the average F1-score across the three classes is 87%. This technique is 400 to 600 times more efficient than Nuclear Medicine residents.

Guo et al. [4] with the ResNet18 model demonstrated that we can utilize this model for categorizing and diagnosing residual thyroid tissue in SPECT images (none of the prior studies have provided any insight into this topic). ResNet18 was trained with 335 thyroid SPECT scintigraphy images from the training set, and then the performance of the proposed model was evaluated using 111 thyroid SPECT scintigraphy images from the testing set. Experiment results are excellent with accuracy of 96.69%, sensitivity of 94.75%, specificity of 99.6%, F1-score of 97.55%. Despite they do not publish dataset, it is clear that vast majority of SPECT images in this study have strong contrast between the patient's body and the background as long as the thyroid glance is also apparent.

We developed a method that addresses the shortcoming in mentioned prior works: easy data with strong contrast, no explanation for the impact of image quality on model's predictions. We separated our dataset into easy and difficult examples, and then analyzed experiment results on each distinct dataset. Our solution's workflow is depicted in Fig. 1.

III. DATA AND METHOD

The most frequent therapies for thyroid cancer are surgical removal and radioactive iodine (RAI) therapy [5]. In patients undergoing RAI therapy, SPECT scan has been an essential imaging modality for risk

stratification, assessment of residual disease, therapeutic planning, and detection of recurrence. SPECT (Single Photon Emission Computed Tomography) is a method of computed tomography using a single photon emission isotope using several radioisotopes to the patient. This imaging technique was first used in the 1960s, before the introduction of computed tomography (CT). With the enhancement of the gamma camera's homogeneity of field of vision, the single photon emission isotope imaging technique has advanced significantly. In nuclear medicine, SPECT has become one of the typical imaging techniques.

We gathered 1,777 SPECT pictures of thyroid cancer patients diagnosed with low dose ^{131}I of 2 mCi or 5 mCi after total thyroidectomy. Doctors use these pictures to diagnose whether thyroid tissue remains in patient's body. There were 864 instances of increased radioactivity, 913 normal instances in the neck region. The final diagnosis were obtained after consultation amongst three physicians with over ten years of expertise in nuclear medicine imaging. All patients with no remaining thyroid tissue will have blood testing to check proper function and clinical parameters.

We refer to American Thyroid Association's guidelines (ATA guidelines) for identifying patients as having a high, moderate, or low recurrence risk. We utilize the clinical outcomes of the research participants as the benchmark to determine the number of individuals with high, medium, and low risk regions in this study.

We broke down dataset by gender, thus there are 352/1,777 instances of men accounting for 19.8%, and 1,425/1,777 instances of women accounting for 80.2%. All pictures were captured by GE Healthcare's INFANIA brand - the most advanced SPECT system available today, and processed with Xelaris software. The methodology for image capture employs a parallel hole collimator with energies between 40 keV to 511 keV, and 9.5mm of crystal thickness. The characteristics of the research individuals are displayed in the TABLE I.

A. Preprocessing

Fig. 2 depicts original SPECT scintigraphy pictures, which should not be used. First, it is evident from Fig. 2 that in addition to the thyroid, other irrelevant physiological organs also take ^{131}I . Second, there are background ghosts and edge artifacts in the source photos, which impact the early stages of the feature extraction process.

TABLE I: Character of study subjects

Parameters	Value
Mean age (years)	44.5 ± 5.5
Range	22 ~ 62
Female rate	1425 (80.2%)
Male rate	352 (19.8%)

We discovered that clinicians rely not just on SPECT scintigraphy pictures, but also on other clinical patient data such as TSH, Thyroglobulin, and TSH (TG). However, extra clinical information is required only in hard instances that leave clinicians perplexed when relying just on SPECT pictures for diagnosis. In these difficult circumstances, clinicians frequently misidentify the thyroid bed as long as it has been absorbed. Fig. 3 depicts the SPECT photography process.

Crop: Each image with dimension 1179×100 pixels in original SPECT scintigraphy images is compressed using JPEG format. We divide our dataset into 2 groups as Fig. 2 while non-suspected group (easy example) contains cases which need only SPECT scintigraphy for diagnosing, its instances consist of 2 whole scan body (WSB): anterior (ANT) and posterior (POST). On the other hand, the suspected group (hard example) need additional clinical information for diagnosing and head and neck (HN) SPECT scintigraphy image. ANT, POST and HN components are separated from parent image following 4-consecutive-step process:

Each original SPECT scintigraphy image with dimensions of 1179×100 pixels is compressed using the JPEG format. While the non-suspected group (easy example) consists of situations for which SPECT scintigraphy is sufficient for diagnosis, its instances comprise of two whole scan bodies (WSB): anterior (ANT) and posterior (POS). In contrast, the suspected group (hard example) requires further clinical information and a head and neck (HN) SPECT scintigraphy imaging for diagnosis. ANT, POST, and HN components are extracted from the original image using following 4-step procedure:

- 1) Convert parent image to grayscale, then complement it by subtracting each pixel value from 255.
- 2) Apply a threshold value to the previous picture to obtain the vertical and horizontal segment lines that divide the ANT, POST, and HN components. We set upon $threshold = 128$.
- 3) Generate mask image matching to frame of ANT, POST, HN based on head, end node of vertical, horizontal segment line, and then apply these masks to the original image to crop required elements.
- 4) Cropped head and neck area with ANT, POST portion to eliminate extraneous physiological organs such as salivary glance, viscera.

Histogram equalization: Histogram Equalization (HE) techniques help to enhance the image so that it gives an improved visual quality and a well defined problem [6]. We used HE to boost the contrast quality of our cropped SPECT scintigraphy images. HE is a crucial pre-processing step that clarifies the absorbed region of the thyroid bed; without it, the model will focus erroneously on the thyroid bed if the patient's

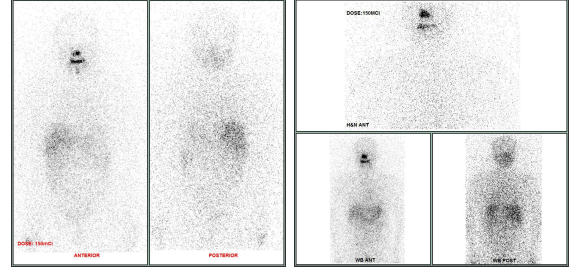


Fig. 2: A representative sampling of each dataset. Left: An example of the nonsuspected group for which anterior and posterior body views are sufficient for diagnosis. Right: An instance of a suspected group or tricky situation for which we require an additional head and neck SPECT scintigraphy.

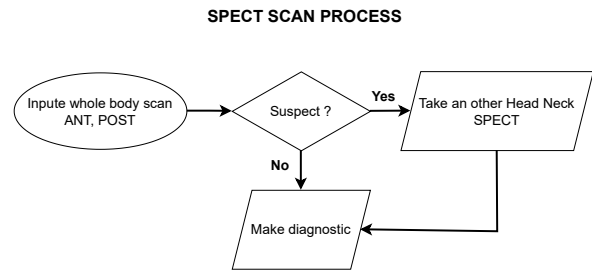


Fig. 3: The process of realizing SPECT scintigraphy.

border cannot be distinguished from the background. HE is motivated by the fact that the thyroid gland may be found based on the position of the shoulders and neck on the body border, which is only achievable when the body boundary is discernible. Fig 4 shows the effect of histogram for enhancing quality of SPECT scintigraphy image.

B. Data Augmentation

Data Augmentation (DA) is a technique used to address the problem of limited data. DA directly synthesizes data using numerous changes that preserve the image's essential characteristics. Diverging source image into numerous manifold neighborhood instances increases model generalization while also increasing local invariance prior. In our work, we employ scale augmentation because it is suitable for SPECT scintigraphy, but other methods, such as shift, rotation, and flip, produce unsatisfactory standard scintigraphy.

C. Deep learning models

Deep learning, in contrast to early machine learning algorithms, is able to automatically extract features as opposed to manually searching for the best ones. Convolution neural network (CNN) has made a significant contribution to the development of deep learning and produced remarkable results in computer vision. Researchers have successfully applied CNNs for many medical image understanding applications

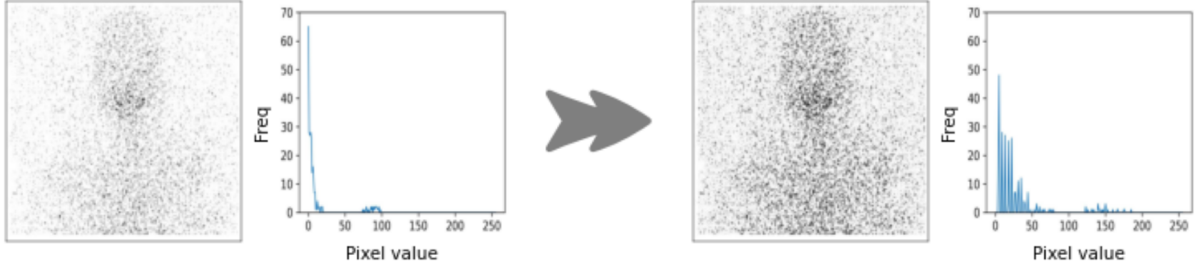


Fig. 4: The impact of histogram equalization on SPECT scintigraphy improvement. Left: SPECT images and the corresponding histogram before to applying the histogram equation. Right: the identical content as the left. It is evident that Histogram equalization increases contrast by creating more homogeneous histogram.

like detection of tumors and their classification into benign and malignant [7], detection of skin lesions [8], detection of colon cancer [9], blood cancer, anomalies of the heart [10], breast [11], chest, eye etc [12]. In this section, we will examine the usual architectures used to analyse and interpret medical images such as GoogLeNet [13], Resnet50 [14], DenseNet [15], VGG16 [16], EfficientNet [17], and Inception ResNet V2 [18].

All mentioned models performed exceptionally well on ImageNet for image classification and detection. We shall demonstrate its effectiveness using SPECT scintigraphy images. Due to the restricted number of SPECT scintigraphy photos (2000), it is extremely challenging to optimize a deep learning model trained from scratch with such a tiny amount of data. Especially when large models are trained on sparse datasets, they are prone to overfitting. Transfer learning is our solution in this case; learned weights from a previously-trained model serve as a good initialization for derived tasks, and it significantly accelerates the training process. On our dataset consisting of thyroid SPECT scans as input and residual or non-residual diagnoses as target labels, we fine-tuned a pre-trained deep learning model and compare the performance of above models.

D. Evaluation metrics

This article explores a binary classification problem. It concerns whether thyroid tissue remains in patient body base on thyroid SPECT scintigraphy image. We select accuracy (ACC), positive prediction value (PPV), sensitivity (SEN), specificity (SPEC), F1 and area under curve (AUC) to evaluate the performance for several models. The model with the highest AUC will be choice as proposed model.

IV. RESULTS

A. Dataset division

Our dataset is divided into two subsets: ANT-HN-POST and ANT-POST. ANT-HN-POST includes only hard samples for which physicians require extra clinical information for accurate diagnosis. ANT-POST includes both difficult and simple samples that

clinicians can diagnose using WB ANT and WB POST. Instead of grouping all simple samples into the ANT-POST set, we also padded difficult samples because the number of simple cases in the original dataset was limited. TABLE II provide comprehensive statistical data for two subsets.

In order to evaluate the generalization of models, we additionally split a validation comprising 25% of training data across ANT-HN-POST and ANT-POST. We established a 2-channel image consisting of pre-processed WSB ANT and pre-processed WSB POST as the input of experimental models for both of the subdatasets mentioned above. All inputs are then resized to 256×256 , the standard dimension of the specified deep learning model.

B. Fine-tuning deep learning model

We independently fine-tuned the pre-trained deep learning model mentioned in Section III-C using our two subdatasets. This strategy will accelerate training by utilizing the trained hidden unit of these models in the ImageNet dataset. Our classification topic differs slightly from ImageNet, and we must make adjustments. As the ImageNet dataset contains RGB images with three channels (Red, Green, Blue), it is necessary to replace the first 3 channels convolution layer of the experimental model with one with only two channels. ImageNet is a 1000-classification task, which necessitates the replacement of the last fully connected layer for our binary classification.

As seen from TABLE II, the ANT-HN-POST dataset is imbalanced and significantly skewed toward

TABLE II: Train the portion of the ANT-POST dataset where suspected instances predominate, in addition to additional non-suspected cases. ANT-HN-POST only has cases of suspicion.

Dataset	Train/Test	#Samples	#Positive	#Negative
ANT-POST	Train	1575	764	811
	Test	200	99	101
ANT-HN-POST	Train	1028	296	732
	Test	100	50	50

the non-residual class. Another interesting discovery from TABLE II is that the ANT-POST subset is dominated by hard samples. These important observations suggest that focal loss is a suitable loss function [19].

We use ADAM optimizer during training process which based on adaptive estimates of lower-order moments, it is computationally efficient, has little memory requirements as long as is well suited for large parameters [19]. TABLE III displays setting of our training parameters.

C. Experiment results

The experimental results are summarized in TABLE IV. We begin by analyzing findings from the ANT-POST dataset, which contains both simple and complex cases. ResNet50 is the most stable model with consistent evaluation measures. Specifically, ResNet50 achieves the highest AUC, which is the aim metric for this study, hence ResNet50 was chosen as the proposed ANT-POST set. Our proposed model has an accuracy of 87%, a sensitivity of 84.85%, a precision residual class of 88.42%, a specificity of 89.11%, an average F1-score of 87%, and an AUC of 0.93. Although the ANT-POST dataset contains a significant number of difficult cases, as seen in TABLE II, our proposed model still performs admirably, demonstrating that balancing difficult with easy examples improves classification performance. These results demonstrated that transfer learning is a reliable approach for automatically diagnosing thyroid SPECT scintigraphy.

Then, we consider a more challenging experiment when using models to diagnose on ANT-HN-POST set containing only hard cases without additional clinical information but only using thyroid SCPECT scintigraphy. Clearly to see that ResNet50 outperforms other models all most metrics including target metric (AUC) thus ResNet50 is our proposed model for dealing with hard examples. The classification accuracy of proposed model is 74%, the sensitivity is 76%, the precision is 73.08% on "residual" class, the specificity is 72%, average F1-score is 74% and AUC is 0.79. These assessment indicators demonstrate that the proposed model is more stable and less biased than those of other models.

TABLE III: Training parameter configurations

Parameters	Dataset	
	ANT-POST	ANT-HN-POST
Batch size	32	32
Patient	10	10
Epoch	20	20
γ	1	0
$[\alpha_0, \alpha_1]$	[2, 3]	[2, 3]
Learning rate	10^{-4}	10^{-4}

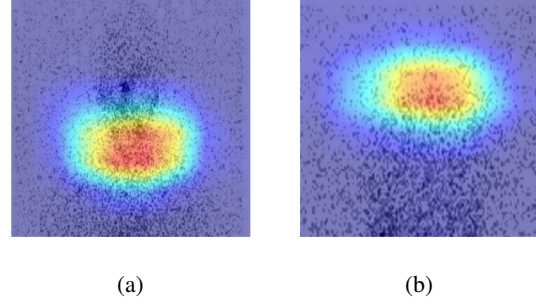


Fig. 5: Model's attention areas that allowed model to predict residuals. (a) is true negative case with confidence 99.99%. (b) are true positive cases with confidence 98.95%.

This section concludes with the discussion of an interesting experiment. We sample from incorrect cases produced by ResNet50 in ANT-HN-POST dataset and prepare a few samples for testing. To explain model prediction, we generate heat map showing model's focusing region for images in Fig. 5. We can see from Fig. 5 that the model paid attention to the thyroid bed region with high confidence.

V. CONCLUSION

We developed a new technique employing thyroid SPECT scintigraphy to determine if any thyroid tissue remains in the patient's body after a total thyroidectomy. In the initial stage, the original SPECT images were pre-processed by clipping irrelevant physiological organs, excluding the head and neck region, in order to improve image contrast using histogram equalization. Then, we divided the obtained data into two subsets, with ANT-HN-POST containing only hard samples and ANT-POST containing both easy and hard samples. We then fine-tuned pre-trained models on each subset, including ResNet50, VGG16, GoogLeNet, DenseNet, Inception ResNet V2, and EfficientNet, and evaluated their performance based on the AUC metric.

TABLE IV: Evaluate performance of various models with ANT-HN-POST and ANT-POST dataset

Dataset	Model	ACC	SEN	PPV	SPEC	F1	AUC
ANT-POST	VGG16	50.50	0	0	100	33.55	0.5
	ResNet50	87	84.85	88.42	89.11	87	0.93
	EfficientNet	80	76.77	81.72	83.17	79.97	0.90
	DenseNet	87	84.85	88.42	89.11	87	0.91
	IncResnet V2	84	70.71	95.89	97.03	83.68	0.89
	GoogleNet	88	82.83	92.13	93.07	87.96	0.92
ANT-HN-POST	VGG16	50	0	0	100	33.33	0.5
	ResNet50	74	76	73.08	72	74	0.79
	EfficientNet	50	0	0	100	33.33	0.48
	DenseNet	70	62	73.81	78	69.81	0.75
	IncResnet V2	61	26	86.67	96	55.55	0.68
	GoogleNet	59	38	65.52	80	57.11	0.64

Using thyroid SPECT scintigraphy pictures, all experiment results in our study indicate that transfer learning using deep learning models are capable of diagnosing if thyroid tissue remains in the body. Our proposed model can perform well in most situations; nevertheless, there are a few challenging circumstances in which the position of the thyroid bed cannot be determined due to poor contrast and blur absorbed thyroid bed. Our experiment shown that the proposed model is more stable, and that we may greatly enhance classification accuracy by balancing difficult with simple samples. When the quantity of data for certain hard examples grows and becomes more balanced, we will conduct experiments using hard examples that are balanced with hard examples. We hope it will be superior to simple samples.

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